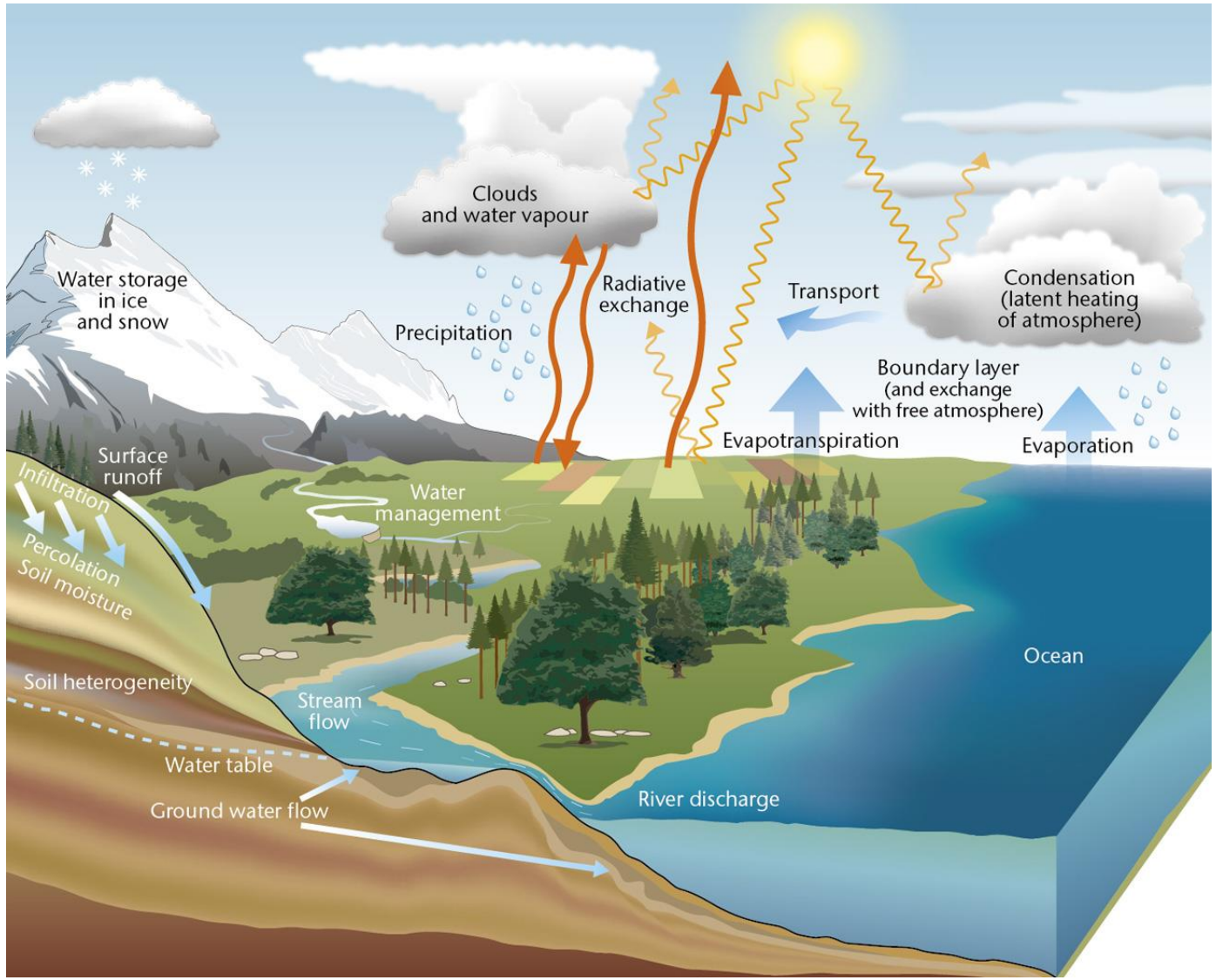
Can machine learning help in reconstructing hydrometeorological time series from the past?

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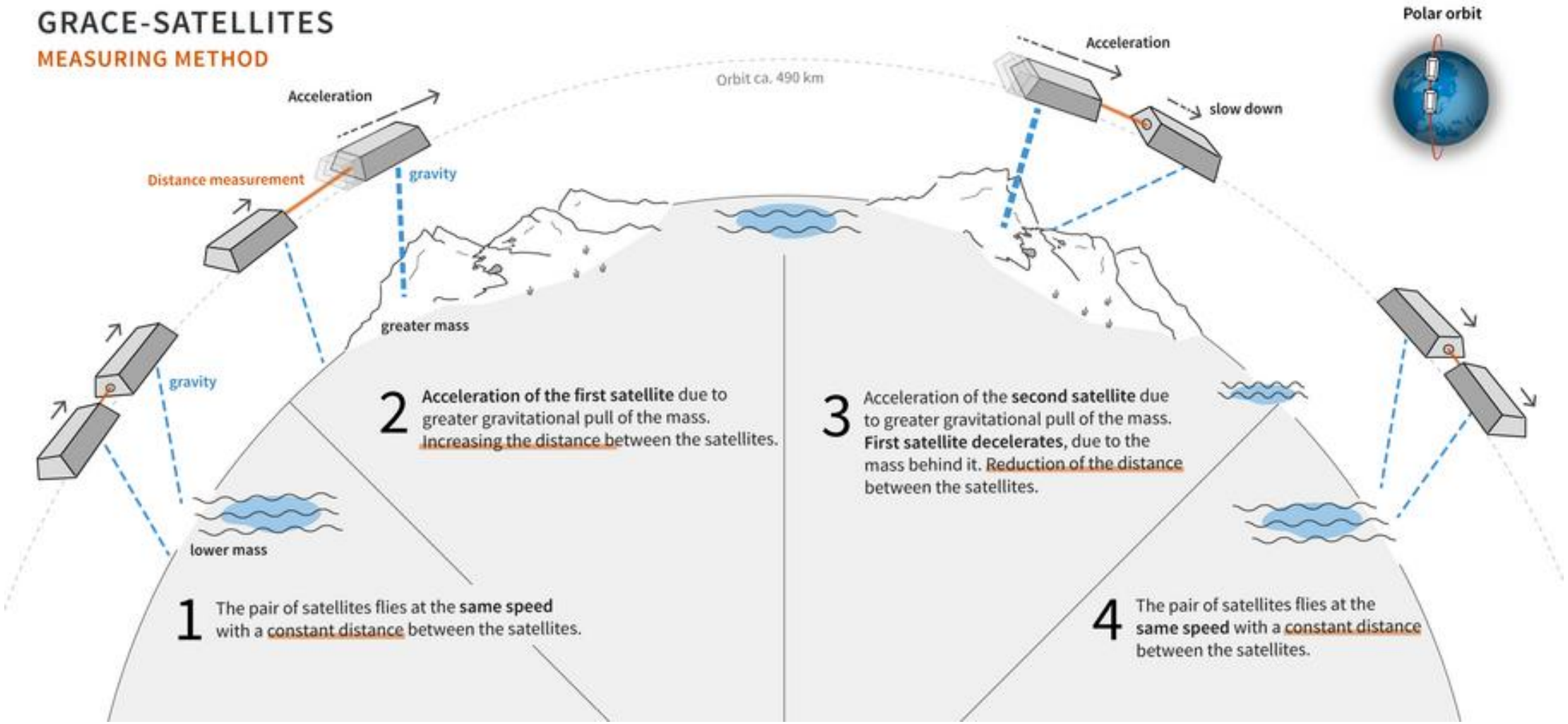
Geodesy for Earth System Science @ IISc



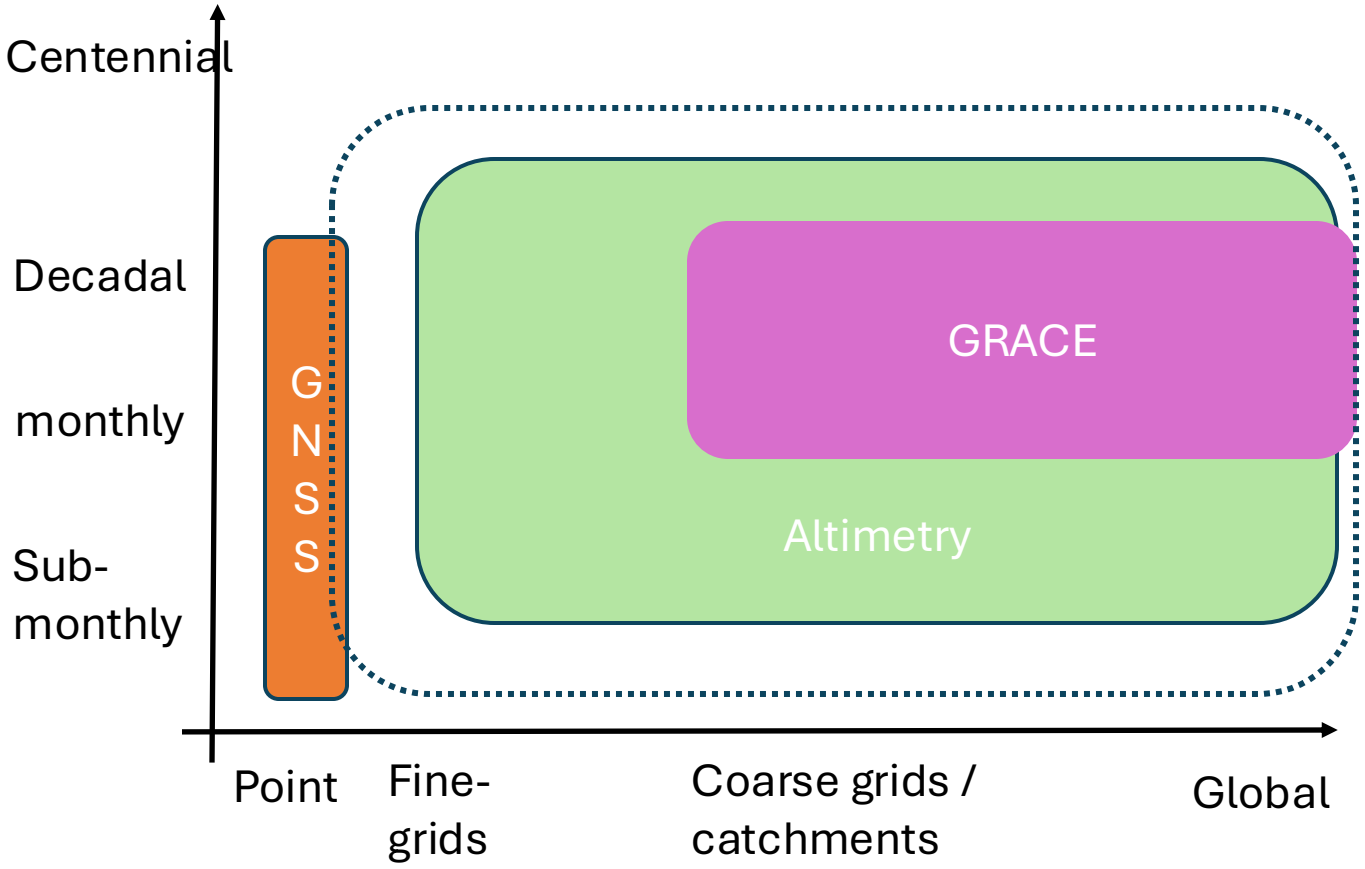
Study changing hydrosphere:
Land hydrology, sea level rise, glacier and ice-sheet mass balance

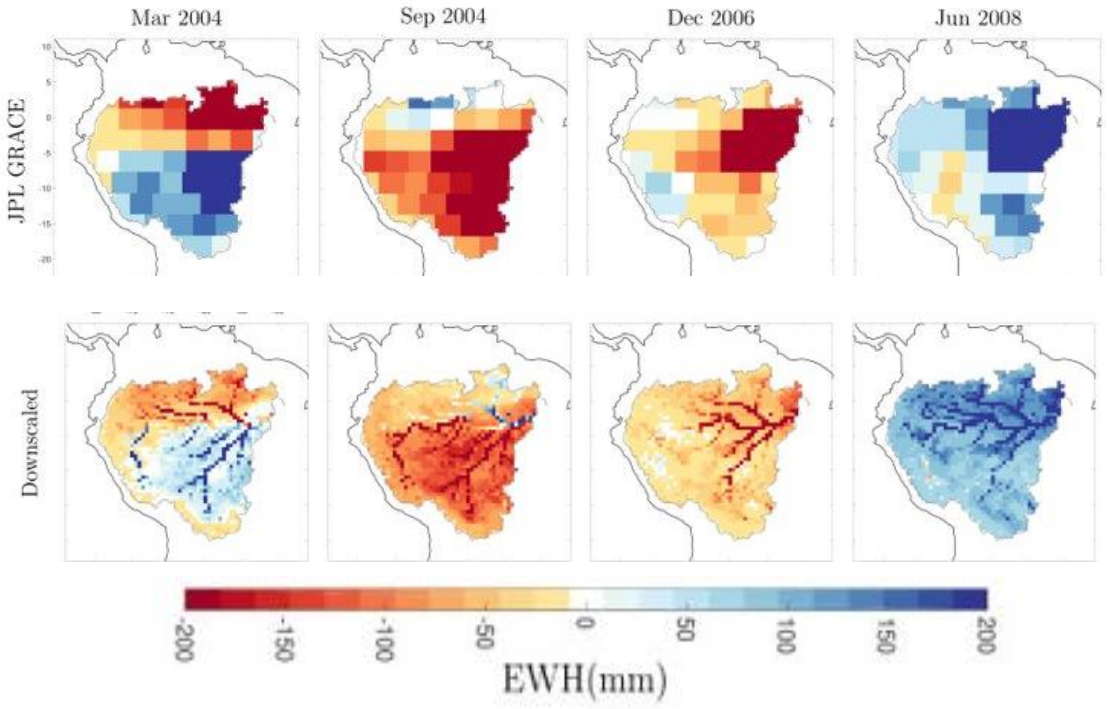
- Satellite Geodesy : GRACE, Altimetry, GNSS
- Remote sensing products
- Physics-based models
- In-situ data collection

GRACE satellites



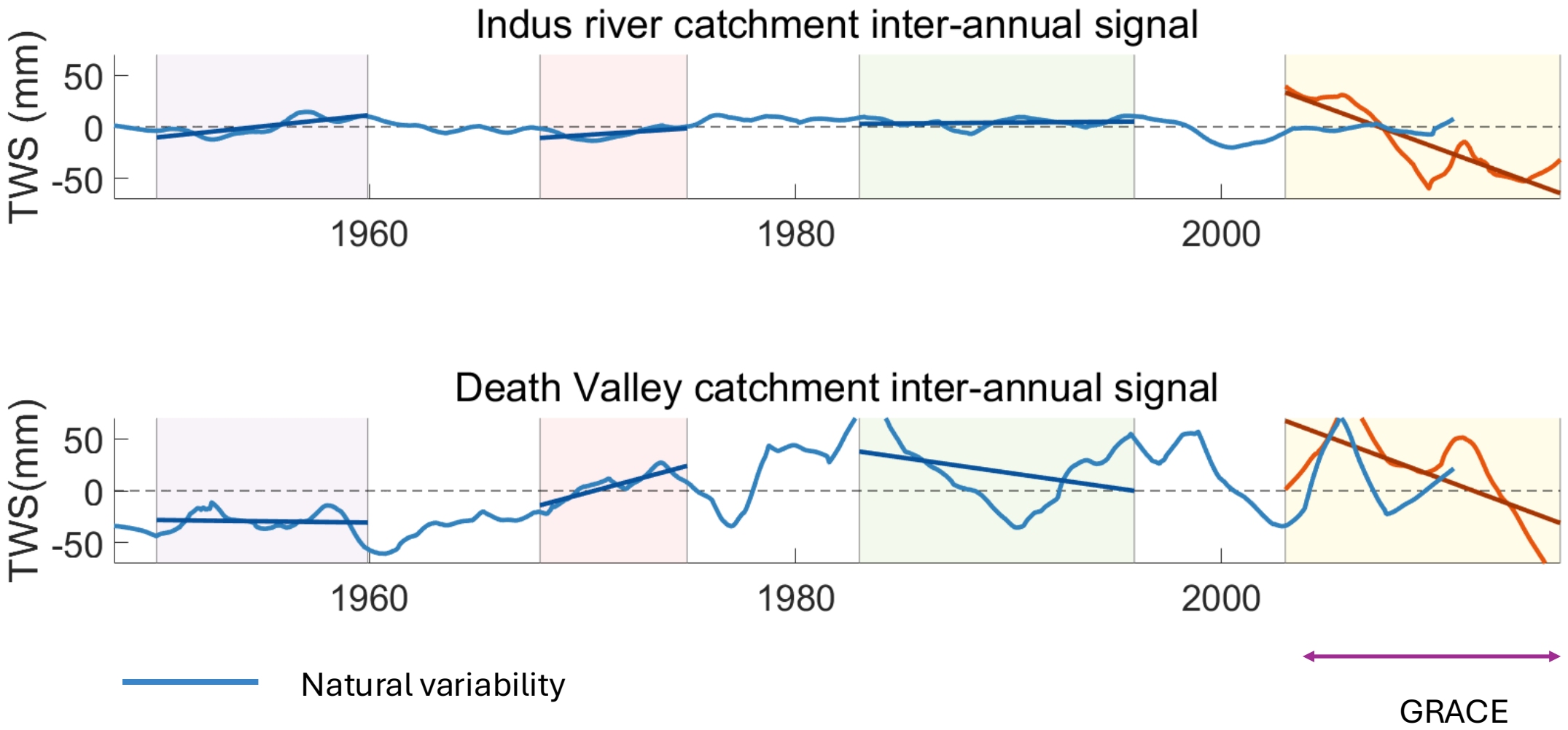
Limited scales





GRACE data processing, Altimetry data processing, Data assimilation, Kalman filtering

Understanding drivers of change



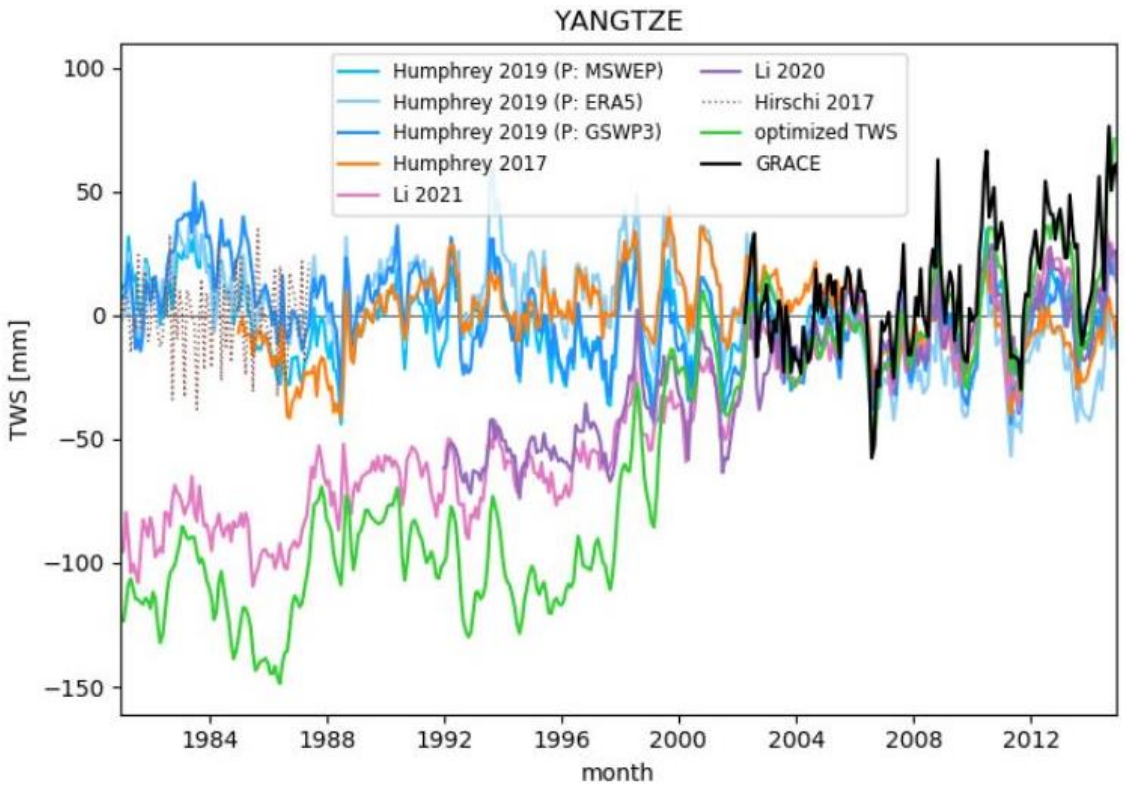
Hence reconstruction!

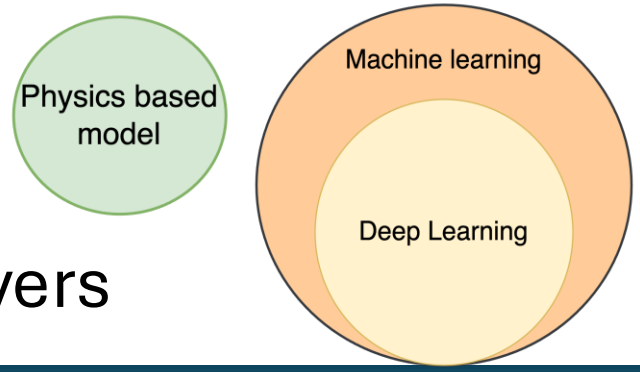
1. Can help in realizing past and enable climate-analysis.

2. Can generate data over contemporary periods with improved coverage: both in space and time.

Usual approach for reconstructing TWS

Authors	Predictors	Model
Senviratne et al., 2004, 2016	-	GCM
Reager et al., 2013	P, T, Forest Cover, Soil water holding capacity	Transfer function, Fourier Analysis
Humphrey et al., 2017	P, T	MLR
Humphrey & Gudmundsson al., 2019	P, T	MLR
Li et al., 2019	P, T, SST, Climate Indices	CNN
Forootan et al., 2020	-	-
Sun et al., 2020	P, T, NOAH-TWS	MLR, DNN, SARIMAX
Liu et al., 2021	P, T	MLR
Chandanpurkar et al., 2022	P, T	Cyclostationary Empirical Orthogonal Functions
Yin et al., 2023	ERA5 model output, Solar-induced chlorophyll fluorescence;	ANN, RF, GLM, GAM, MARSs
Palazzoli et al., 2025	ERA5 model output, Solar-induced fluorescence	LSTM, Bi-LSTM
Mandal et., 2025	P, T, NAOH-TWS, CLSM-TWS, Climate Indices	CNN, SVR, SER, ETR
Genter et al., 2026	ERA5 model output, SST, ISIMIP Land Use, ISIMIP Lakes fraction	CNN-LSTM
Saemian et., 2026	13 LSM & GHM TWS	RF



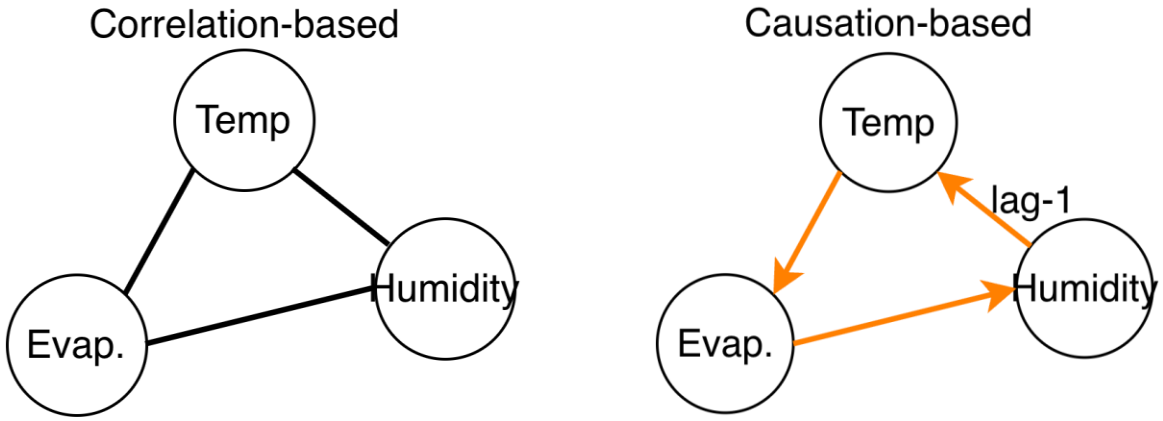


Correlation is the primary tool to identify drivers

Correlation is not causation

- Causal Discovery (CD)
- Discovering a direct cause-effect relation between entities (variables or events).

- Causal Inference
- Quantifying the influence of an entity (variable or event) onto another.



- Structure
- Directionality

Runge et al., 2019

Causal discovery

- Granger Causality
- Noise-based
- Constraint-based
- Score-based
- State-space reconstruction based: Convergent Cross Mapping
- Others

TCDF (Nauta et al., 2020)

- Find DAG

- Method: CNN to model time-series, Attention scores to unravel causality

VARLiNGAM (Hyvärinen et al., 2008)

- Classic SVAR model

- Method: Exploits lower triangular ordering of adjacency matrix

PCMCI+ (Runge et al., 2022)

- Find DAG

- Method: Conditional independence tests

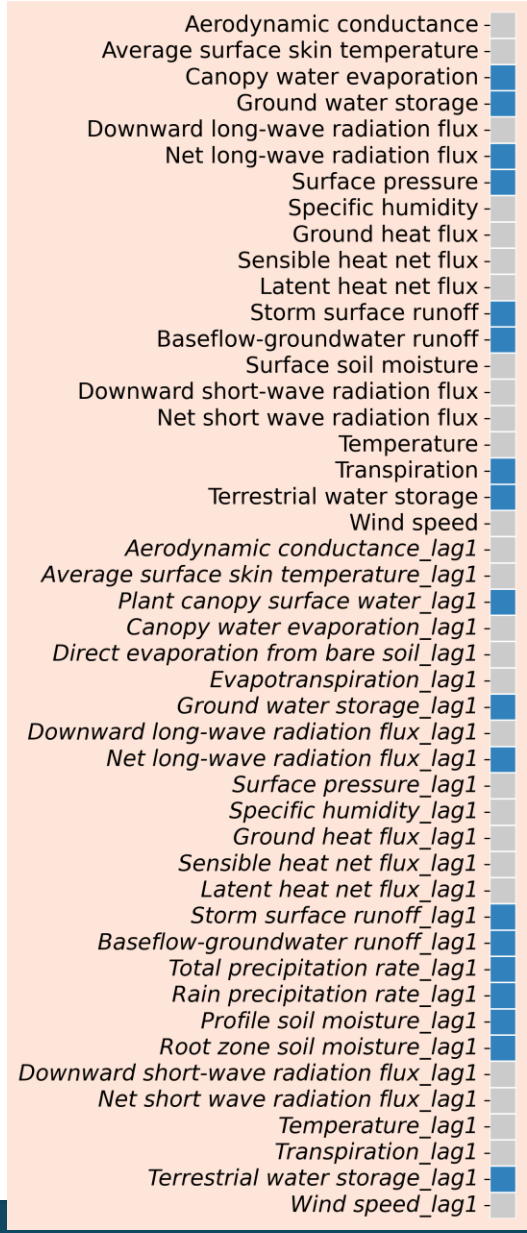
DYNOTEARS (Pamfil et al., 2020)

- Classic SVAR model

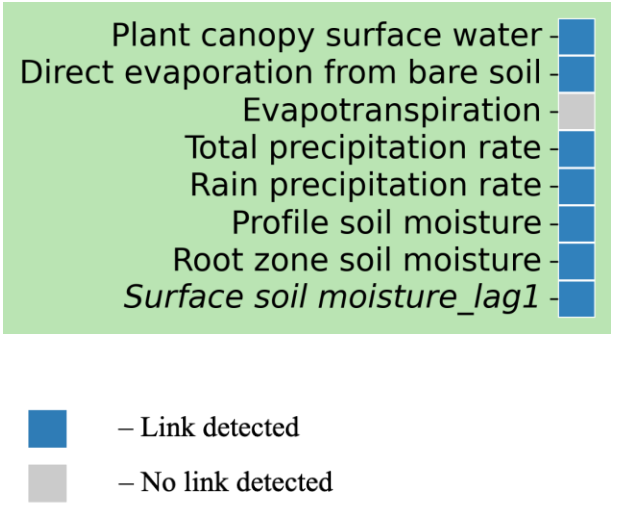
- Method: Numerical optimization to minimize the model error and acyclicity score

Causal links in a hydrometeorological system?

- Hydrometeorological systems are highly correlated:
  - multiple cross and lagged correlations, typical to hydrology.
- Correlation does not imply causation.
- *getting right answers for wrong reasons*.



Correlation of surface soil moisture in GLDAS model environment



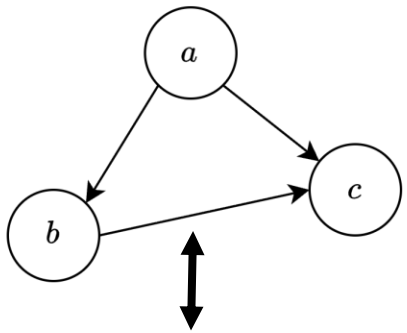
Causal discovery of links: synthetic environment

Node: Variable (timeseries) or event

Link/Edge/Adjacency: Relationship b/w nodes

Directed edge: Cause effect in direction of edge

Directed-Acyclic Graph (DAG)



Adjacency Matrix

	a	b	c
a	0	1	1
b	0	0	1
c	0	0	0

$$X_t^1 = \eta_t^1$$
$$X_t^2 = a \cdot X_t^1$$
$$X_t^3 = b \cdot X_t^1 - c \cdot X_t^2 + d \cdot X_{t-1}^1$$

Lower triangular ordering

	X_t^1	X_t^2	X_t^3	X_{t-1}^1
X_t^1	0	1	1	0
X_t^2	0	0	1	0
X_t^3	0	0	0	0
X_{t-1}^1	0	0	1	0

	X_t^3	X_t^2	X_t^1	X_{t-1}^1
X_t^3	0	0	0	0
X_t^2	1	0	0	0
X_t^1	1	1	0	0
X_{t-1}^1	1	0	0	0

Recall (*TPR*): ability to accurately identify the correct links in the adjacency matrix

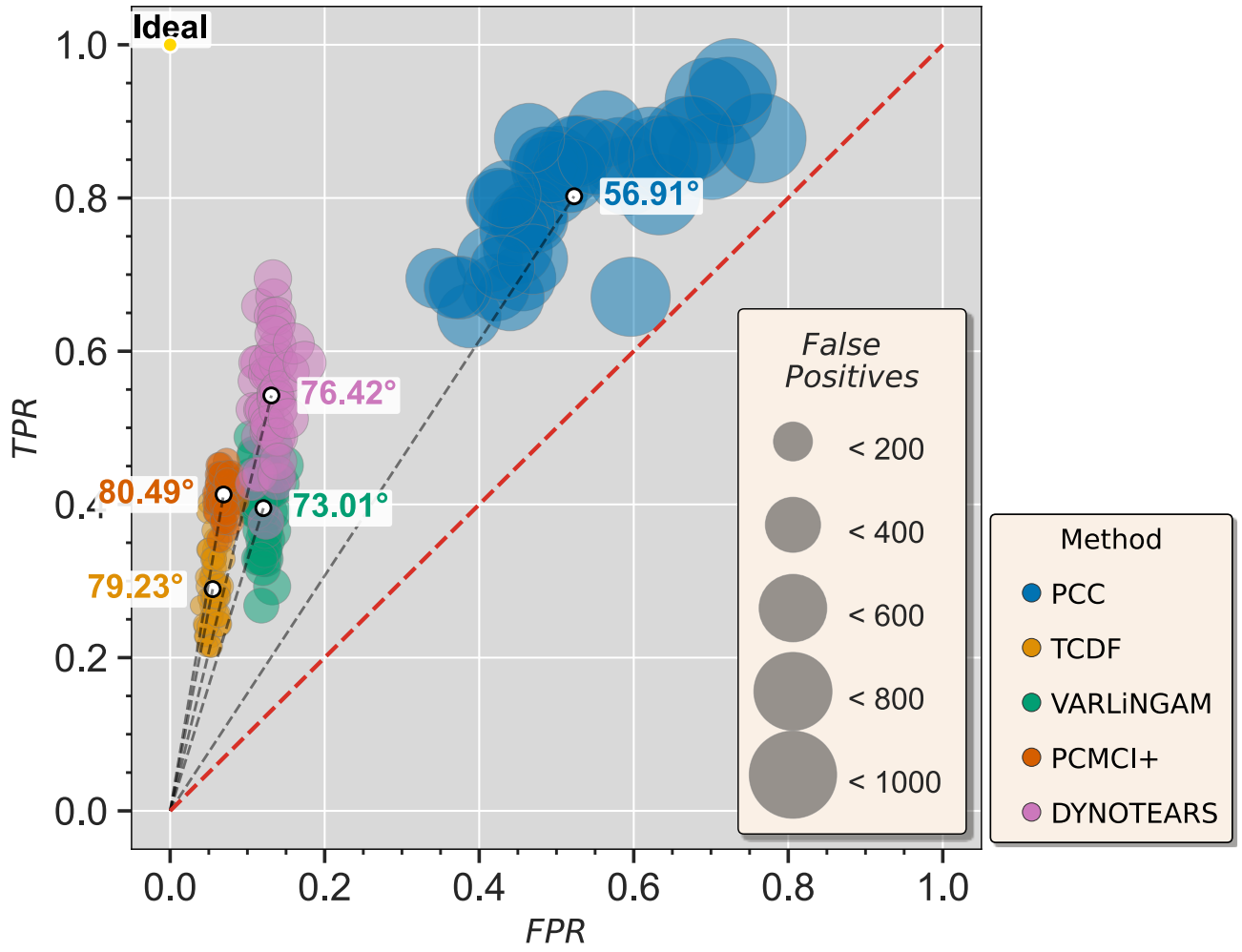
$$= \frac{True\ Positive}{True\ Positive + False\ Negative}, \in [0, 1]$$

False positive ratio (*FPR*): ability to accurately identify the incorrect links in the adjacency matrix

$$= \frac{False\ Positive}{False\ Positive + True\ Negative}, \in [0, 1]$$

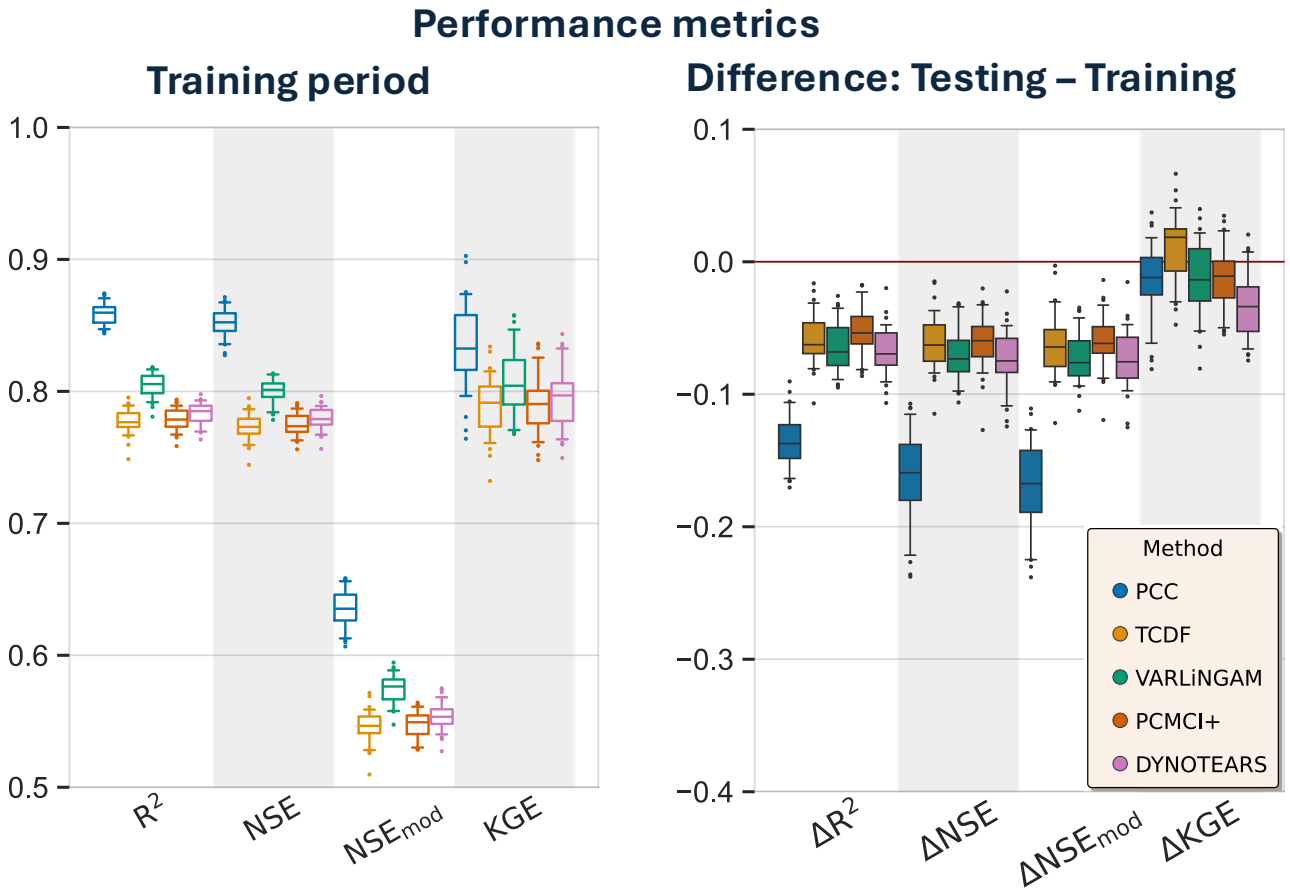
Causal discovery identifies drivers better

- Cost of identifying causal links is the accumulation of false positives.
- PCC shows high variance along TPR and FPR, indicating lack of robustness.
- Comparatively, CD methods show less variance along FPR, demonstrating robustness towards eliminating false positives.
- For a unit gain in FPR, CD methods have a higher TPR.

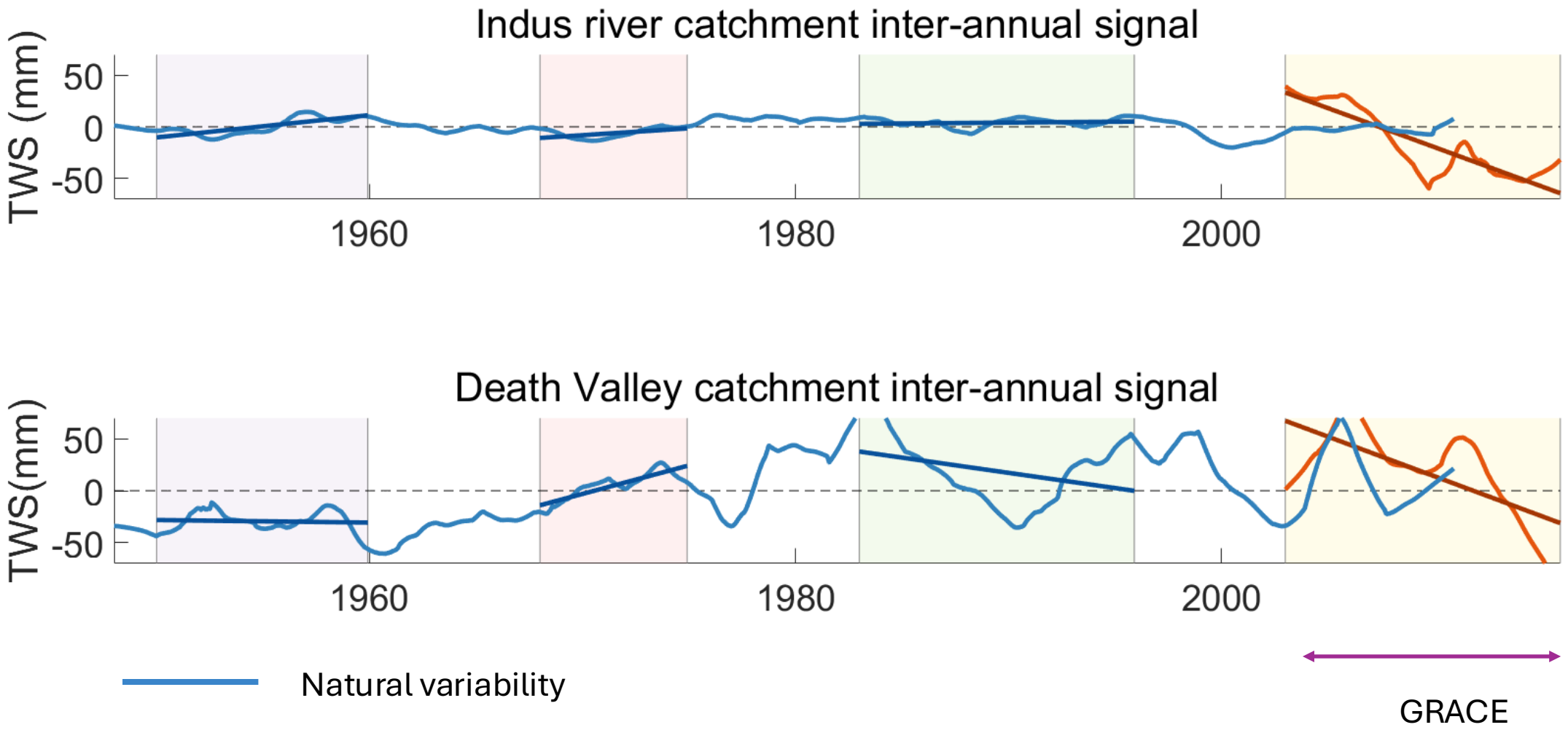


Using CD improves prediction !

Location	Ganga basin, India
Model	Support Vector Regression
Training	Normal period (01 Jan 2000 to 31 December 2003)
Testing	Drought period (01 Jan 2004 to 31 December 2005)
Noise added	Random Gaussian (100 Monte Carlo simulations)



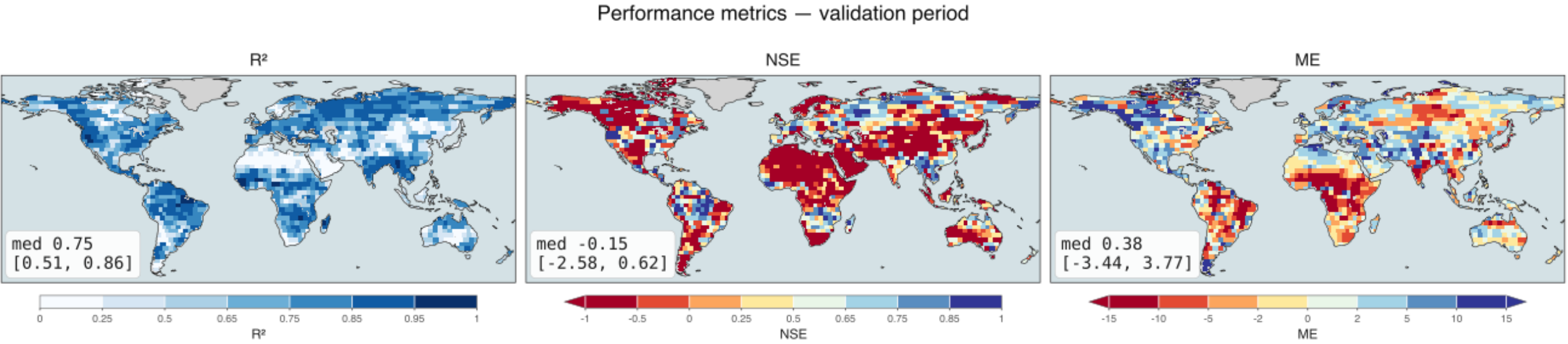
Remember we wanted to know past TWS!



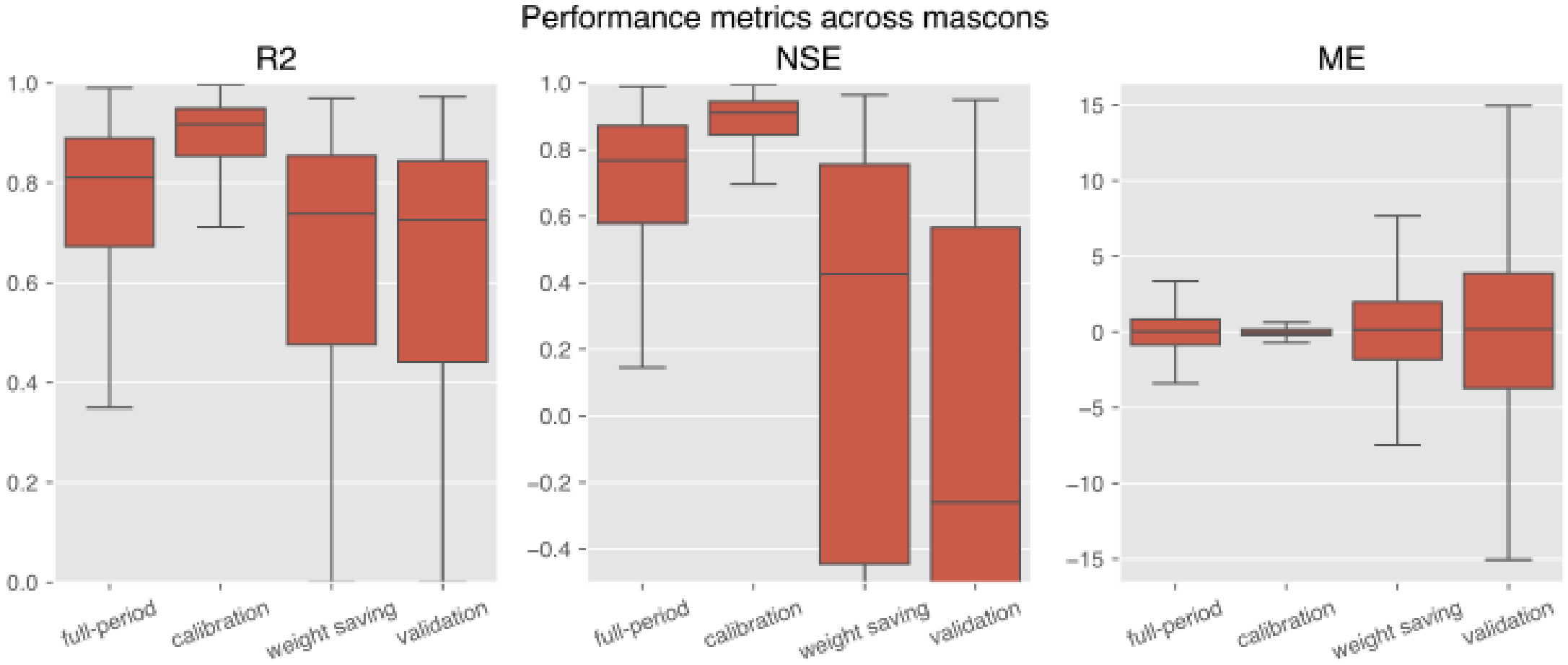
TWS reconstruction

$$S_t = \alpha S_{t-1} + \Delta S_t$$
$$\Delta S_t = NN\ (X_t)$$
$$TWS_t = g(S_t)$$

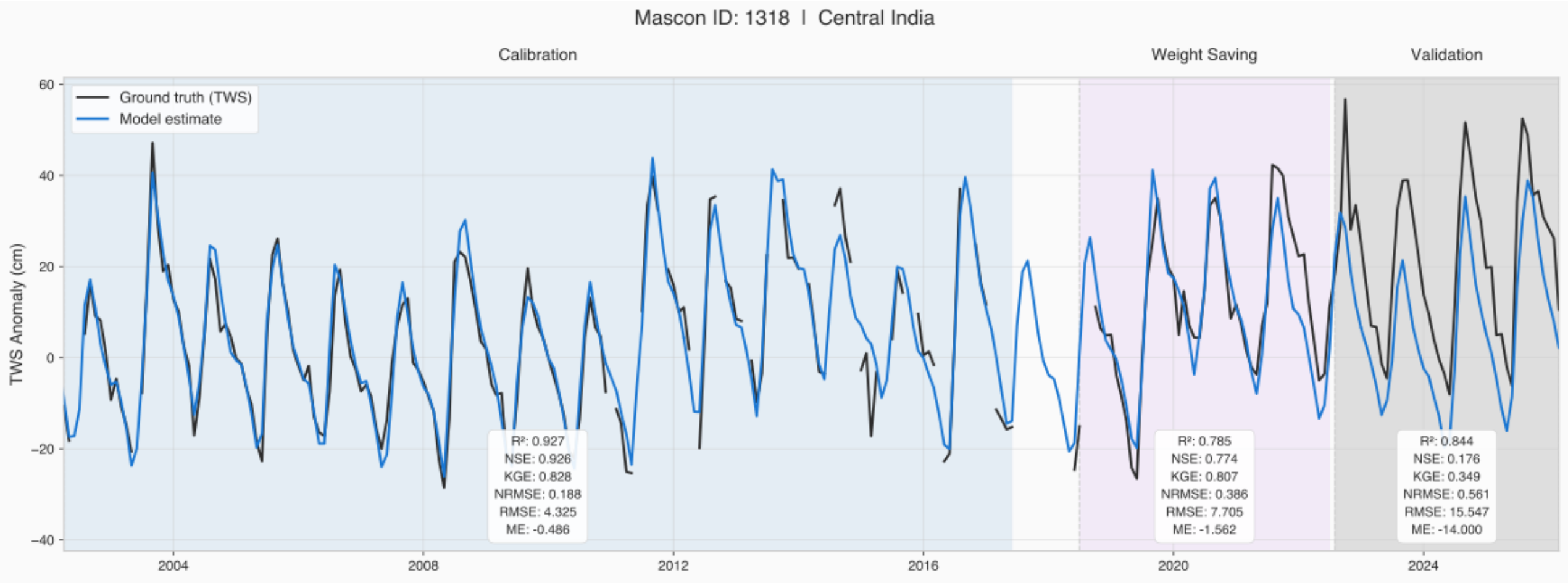
- Here NN (size 32) is a feedforward Neural Network which takes the causal drivers as features.
- The state vector S_t is a vector of size 4, chosen to mimic the four different types of water storages.
- g is a Linear function simply combining the different states together.
- By using an in-direct observation function g allows to exploit the autocorrelated nature of TWS data but also avoid drift induced by explicit autoregressive models (ARMA, ARIMA).



- Figure shows same metrics as the boxplot, but at grid level instead of Mascons (hence the median & IQR difference).



➢This drift is observed globally, resulting in good R2 but poor NSE & ME

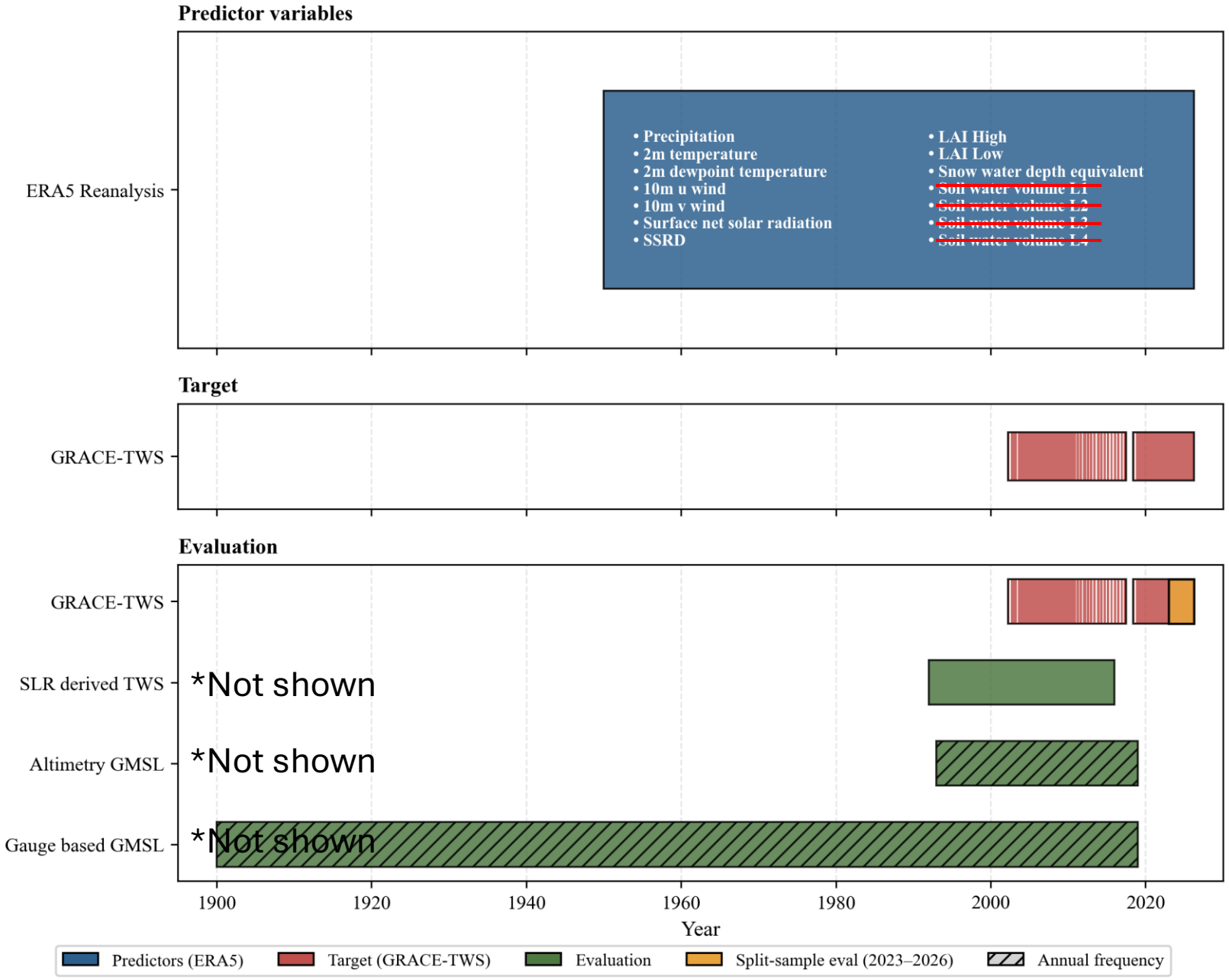

Mascon ID: 1318 | Central India
Calibration
Weight Saving
Validation
Ground truth (TWS)
Model estimate
TWS Anomaly (cm)
60
40
20
0
−20
−40
2004
2008
2012
2016
2020
2024
R²: 0.927
NSE: 0.926
KGE: 0.828
NRMSE: 0.188
RMSE: 4.325
ME: -0.486
R²: 0.785
NSE: 0.774
KGE: 0.807
NRMSE: 0.386
RMSE: 7.705
ME: -1.562
R²: 0.844
NSE: 0.176
KGE: 0.349
NRMSE: 0.561
RMSE: 15.547
ME: -14.000

Open questions and doubts

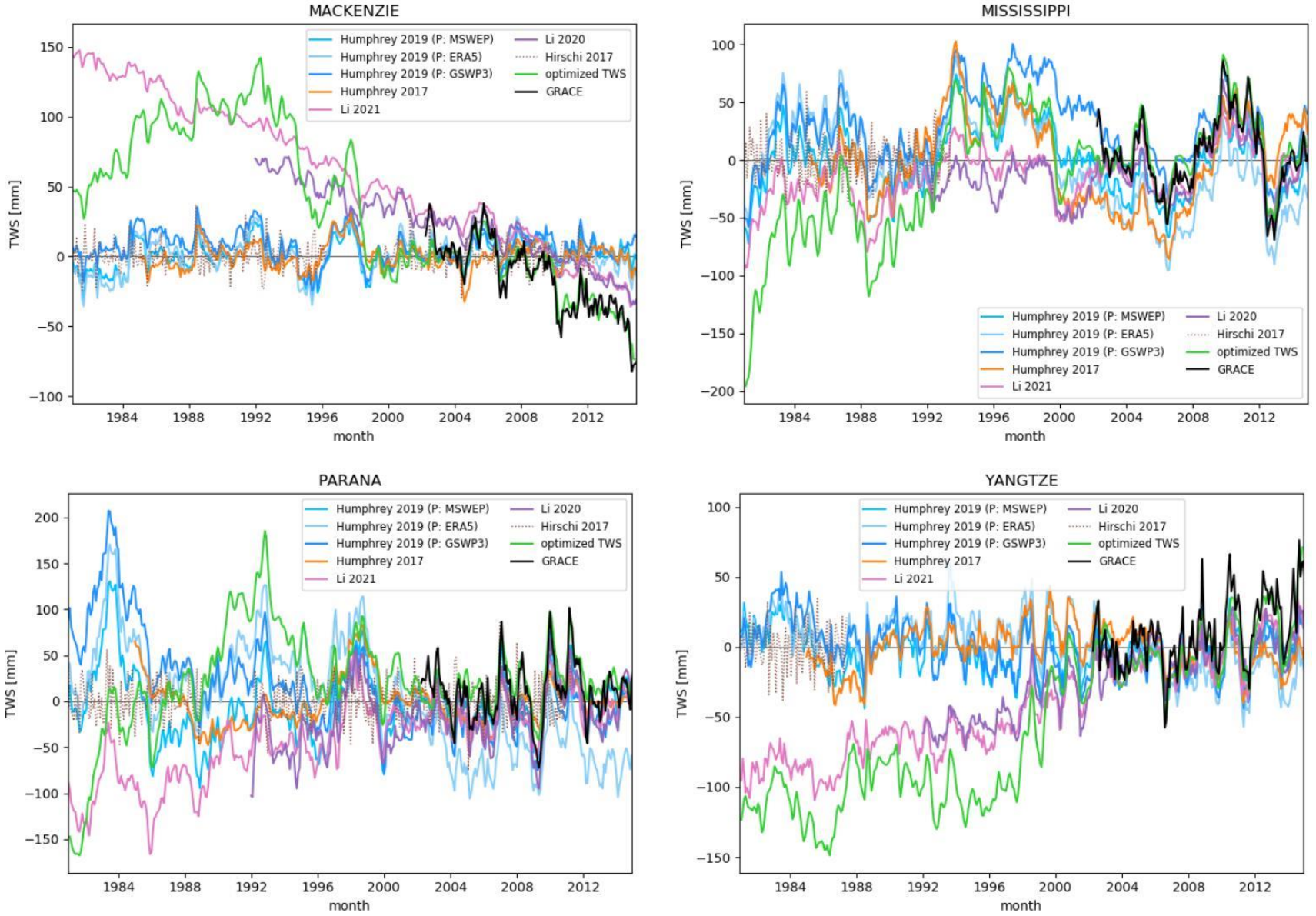
- Capturing non-stationarity!
- Drivers evolve in time and we are not able to cope with this.
- Are there ML models that may prefer causality?

Thanks

Backup slides


Predictor variables
ERA5 Reanalysis
• Precipitation
• 2m temperature
• 2m dewpoint temperature
• 10m u wind
• 10m v wind
• Surface net solar radiation
• SSRD
• LAI High
• LAI Low
• Snow water depth equivalent
• Soil water volume L1
• Soil water volume L2
• Soil water volume L3
• Soil water volume L4
Target
GRACE-TWS
Evaluation
GRACE-TWS
SLR derived TWS
*Not shown
Altimetry GMSL
*Not shown
Gauge based GMSL
*Not shown
1900
1920
1940
1960
1980
2000
2020
Year
Predictors (ERA5)
Target (GRACE-TWS)
Evaluation
Split-sample eval (2023–2026)
Annual frequency

reconstructed TWS comparison



State space model for TWS estimation

$$S_t = \alpha S_{t-1} + \Delta S_t$$
$$\Delta S_t = NN\,(X_t)$$
$$TWS_t = g(S_t)$$

- Here NN (size 32) is a feedforward Neural Network which takes the causal drivers as features.
- The state vector S_t is a vector of size 4, chosen to mimic the four different types of water storages.
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